

Dhaka International University

Department of CSE



Assignment

Assignment name: Design and Implementation of IP Addressing Using VLSM Subnetting

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Submitted By:	Submitted To:
Name: Md Yeasin Bhuiyan Roll:43 Batch:D-91 Reg: CS-D-91-23-124699	Mir Aminul Haque, Lecturer of CSE Department

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Introduction

IP addressing is one of the fundamental concepts of computer networking. Every device connected to a network requires a unique IP address for communication. In a large network, efficient IP address allocation is very important to reduce address wastage and improve network management. Variable Length Subnet Mask (VLSM) is an advanced subnetting technique that allows subnet sizes to be created according to the number of required hosts. In this assignment, VLSM is used to design an efficient IP addressing scheme for the given network topology by selecting an appropriate classful IP address and determining the Network Address (NA), Broadcast Address (BA), and Subnet Mask (SM) for each subnet.

1. You have to solve the IP addressing using subnetting technique.

Answer:

To solve the IP addressing problem, the Variable Length Subnet Masking (VLSM) technique has been used because the given network consists of different LANs with different host requirements. VLSM allocates subnet sizes according to the required number of hosts, which minimizes IP address wastage.

The required host information is shown below.

The required host information of the given network is shown below.

Network	Required Hosts
LAN-4	5000
LAN-2	4096
LAN-5	1096
LAN-3	1024
LAN-1	127
WAN-1 to WAN-5	2 Hosts each

Host Requirement Analysis

	Network	Required Hosts	Required Host Bits	Prefix	Subnet Mask
	LAN-4	5000	13	/19	255.255.224.0
	LAN-2	4096	13	/19	255.255.224.0
	LAN-5	1096	11	/21	255.255.248.0
	LAN-3	1024	11	/21	255.255.248.0
	LAN-1	127	8	/24	255.255.255.0
	WAN-1	2	2	/30	255.255.255.252
	WAN-2	2	2	/30	255.255.255.252
	WAN-3	2	2	/30	255.255.255.252
	WAN-4	2	2	/30	255.255.255.252
	WAN-5	2	2	/30	255.255.255.252

2. Choose an appropriate classful IP address from a particular IP class.

Answer:

The total number of hosts in the given network is calculated first.

Total Hosts

$$= 5000 + 4096 + 1096 + 1024 + 127$$

$$= 11,343 \text{ Hosts}$$

Since Class C supports only 254 hosts, it cannot satisfy the requirement.

Although Class A can support a very large number of hosts, using Class A would waste a huge number of IP addresses.

Therefore, Class B is selected because it provides sufficient address space while minimizing address wastage.

The selected Classful Network is

172.16.0.0/16

Default Subnet Mask

255.255.0.0

This network supports up to 65,534 hosts, which is sufficient for the given topology and also allows future expansion.

The total number of required hosts is calculated below.

Total Host Calculation

Network		Required Hosts
LAN-4	5000	
LAN-2	4096	
LAN-5	1096	
LAN-3	1024	
LAN-1	127	
Total Hosts	11,343	

Since the total host requirement is 11,343, Class B is the most appropriate choice.

Selected Network Information

Item	Value
Selected Class	Class B
Network Address	172.16.0.0
Default Prefix	/16
Default Subnet Mask	255.255.0.0
Maximum Hosts Supported	65,534

3. Efficiently allocate the chosen IP address so that wastage of IP address can be prevented.

Answer:

To minimize IP address wastage, the selected Class B network is allocated using the Largest-to-Smallest VLSM allocation method.

In this method, the subnet with the highest host requirement is assigned first, followed by the remaining networks in descending order.

The allocation process is given below.

VLSM Allocation

Allocation Order	Network	Network Address	Prefix
1	LAN-4	172.16.0.0	/19
2	LAN-2	172.16.32.0	/19
3	LAN-5	172.16.64.0	/21
4	LAN-3	172.16.72.0	/21
5	LAN-1	172.16.80.0	/24
6	WAN-1	172.16.81.0	/30
7	WAN-2	172.16.81.4	/30
8	WAN-3	172.16.81.8	/30
9	WAN-4	172.16.81.12	/30
10	WAN-5	172.16.81.16	/30

Address Range

Network	First Usable IP	Last Usable IP
LAN-4	172.16.0.1	172.16.31.254
LAN-2	172.16.32.1	172.16.63.254
LAN-5	172.16.64.1	172.16.71.254
LAN-3	172.16.72.1	172.16.79.254
LAN-1	172.16.80.1	172.16.80.254
WAN-1	172.16.81.1	172.16.81.2
WAN-2	172.16.81.5	172.16.81.6
WAN-3	172.16.81.9	172.16.81.10
WAN-4	172.16.81.13	172.16.81.14
WAN-5	172.16.81.17	172.16.81.18

After allocating all LAN networks, the remaining address space is assigned to the WAN links using /30 subnet masks.

This allocation method minimizes unused IP addresses and keeps sufficient address space available for future network expansion.

4. In case of IP allocations, derive the Network addresses, Broadcast addresses for each of the subnets along with their subnet masks.

Answer:

After completing the subnet allocation, the Network Address, Broadcast Address and Subnet Mask of every subnet are determined as follows.

The network address, subnet mask, usable host range and broadcast address for every subnet are shown below.

Final IP Addressing Plan

Network	Network Address	Prefix	Subnet Mask	First Usable	Last Usable	Broadcast Address
LAN-4	172.16.0.0	/19	255.255.224.0	172.16.0.1	172.16.31.254	172.16.31.255
LAN-2	172.16.32.0	/19	255.255.224.0	172.16.32.1	172.16.63.254	172.16.63.255
LAN-5	172.16.64.0	/21	255.255.248.0	172.16.64.1	172.16.71.254	172.16.71.255
LAN-3	172.16.72.0	/21	255.255.248.0	172.16.72.1	172.16.79.254	172.16.79.255
LAN-1	172.16.80.0	/24	255.255.255.0	172.16.80.1	172.16.80.254	172.16.80.255
WAN-1	172.16.81.0	/30	255.255.255.252	172.16.81.1	172.16.81.2	172.16.81.3
WAN-2	172.16.81.4	/30	255.255.255.252	172.16.81.5	172.16.81.6	172.16.81.7
WAN-3	172.16.81.8	/30	255.255.255.252	172.16.81.9	172.16.81.10	172.16.81.11
WAN-4	172.16.81.12	/30	255.255.255.252	172.16.81.13	172.16.81.14	172.16.81.15
WAN-5	172.16.81.16	/30	255.255.255.252	172.16.81.17	172.16.81.18	172.16.81.19

Conclusion

This assignment demonstrates the practical implementation of IP Addressing using the Variable Length Subnet Masking (VLSM) technique. A suitable Class B network was selected according to the total host requirement, and the available IP address space was efficiently divided into multiple subnets. By applying the largest-to-smallest subnet allocation method, IP address wastage was minimized while ensuring that each LAN and WAN received sufficient address space. Finally, all required Network Addresses, Subnet Masks, Broadcast Addresses, and Usable IP Ranges were successfully determined. The proposed addressing plan is well-organized, efficient, and flexible enough to support future network growth and management.